POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 6034C | Semester VI | Theory | 4 Credits

Electrification of Residential Buildings

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electrical & Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6034C as Electrification of Residential Buildings in Semester VI for Electrical & Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on Swayam/NPTEL’s Electricity and Electrical Wiring course and polytechnic electrical estimation curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Electrical designs, wiring plans, material estimates and installation methods must be based on approved engineering plans, certified hardware data, current safety standards (NEC/IE rules) and competent professional review.

Verified source note: Verified sources support wiring systems, estimation techniques, domestic installation, and testing procedures. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Electrification of Residential Buildings studies the planning, design and installation of electrical systems in domestic environments. It develops the ability to analyze wiring systems, evaluate electrical loads, prepare estimation and costing for residential projects, and coordinate installation and testing while respecting the technical and safety boundaries of electrical engineering and national building codes.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□.

Prerequisite foundation

Basic electrical engineering, electrical circuits, engineering drawing and safety practices.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the types of wiring systems, accessories and safety standards for residential buildings.
2. Apply estimation and costing techniques to prepare wiring plans and material lists for domestic projects.
3. Analyze the installation requirements for distribution boards, sub-circuits and earthing systems.
4. Apply systematic testing and commissioning procedures to ensure the safety and reliability of domestic wiring.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□□ exam-□ □□□□□ □□□□□□□□ □□□□□□□ □□□□□ □□□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the wiring systems, accessories and safety rules for residential electrification.	Understand
CO2	Prepare electrical layout plans and estimate material costs for given residential buildings.	Apply
CO3	Analyze the installation and earthing requirements for domestic electrical systems.	Apply
CO4	Perform testing and troubleshooting of residential electrical installations.	Apply

Official CO-PO articulation matrix							
CO	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7–PO11**
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Wiring Systems and Safety Standards	Types of wiring: Cleat, Casing-Capping, Conduit (PVC/MS); Wiring accessories: Switches, sockets, lamp holders, ceiling roses; Indian Electricity (IE) rules for domestic wiring; National Electrical Code (NEC) overview.	Approx. 14 hours	CO1	Module 1
2	Estimation and Costing for Domestic Wiring	Preparation of wiring layout plans; Single-line diagrams; Load calculation for lighting and power circuits; Sizing of cables and distribution boards; Estimation of materials and labor costs.	Approx. 16 hours	CO2	Module 2
3	Installation of Distribution and Earthing	Installation of Main Switch and Energy Meter; Distribution Board (DB) arrangement; Sub-circuits: Lighting and Power; Earthing systems: Pipe and Plate earthing; Earth resistance measurement.	Approx. 16 hours	CO3	Module 3
4	Testing, Commissioning and Troubleshooting	Testing of new installations: Insulation resistance, Continuity, Polarity, Earth resistance; Commissioning procedures; Common wiring faults and troubleshooting; Basics of home automation and UPS installation.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Wiring Systems and Safety Standards

Types of wiring: Cleat, Casing-Capping, Conduit (PVC/MS); Wiring accessories: Switches, sockets, lamp holders, ceiling roses; Indian Electricity (IE) rules for domestic wiring; National Electrical Code (NEC) overview.

CO1 · Approx. 14 hours

Estimation and Costing for Domestic Wiring

Preparation of wiring layout plans; Single-line diagrams; Load calculation for lighting and power circuits; Sizing of cables and distribution boards; Estimation of materials and labor costs.

CO2 · Approx. 16 hours

Installation of Distribution and Earthing

Installation of Main Switch and Energy Meter; Distribution Board (DB) arrangement; Sub-circuits: Lighting and Power; Earthing systems: Pipe and Plate earthing; Earth resistance measurement.

CO3 · Approx. 16 hours

Testing, Commissioning and Troubleshooting

Testing of new installations: Insulation resistance, Continuity, Polarity, Earth resistance; Commissioning procedures; Common wiring faults and troubleshooting; Basics of home automation and UPS installation.

CO4 · Approx. 14 hours

MODULE 1

Wiring Systems and Safety Standards

Types of wiring: Cleat, Casing-Capping, Conduit (PVC/MS); Wiring accessories: Switches, sockets, lamp holders, ceiling roses; Indian Electricity (IE) rules for domestic wiring; National Electrical Code (NEC) overview.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

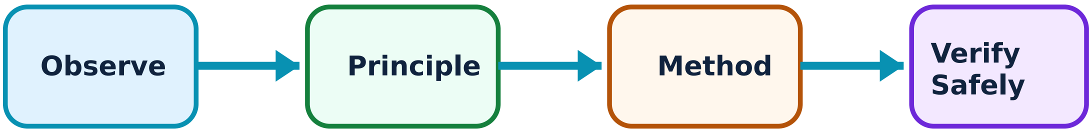
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Wiring Systems and Safety Standards: identify the system, state its principle, follow the correct method and verify the result safely.

1. Wiring systems and accessories−

Electrical wiring involves the systematic installation of cables and accessories. Casing-capping is used for surface wiring, while conduit wiring (surface or concealed) provides better protection. Accessories like modular switches and multi-pin sockets are selected based on current rating and application.

Study and application method

Compare 'Concealed Conduit' and 'Casing-Capping' wiring systems; list five common wiring accessories used in a bedroom.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare 'Concealed Conduit' and 'Casing-Capping' wiring systems; list five common wiring accessories used in a bedroom.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Improper cable selection can lead to overheating and fires. Do not install wiring without authorised current-carrying capacity (ampacity) analysis based on the load.

2. IE Rules and Safety−

The Indian Electricity (IE) rules mandate safety standards for electrical installations. Key rules include minimum height of switches, distance between light points and fans, and mandatory use of protective devices. The National Electrical Code (NEC) provides comprehensive guidelines for residential safety.

Study and application method

State three 'IE Rules' related to domestic electrical installations; explain the importance of 'Electrical Safety Symbols'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State three 'IE Rules' related to domestic electrical installations; explain the importance of 'Electrical Safety Symbols'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Electrical work is hazardous. All domestic installations must follow authorised safety codes and be performed by a certified electrician.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Estimation and Costing for Domestic Wiring

Preparation of wiring layout plans; Single-line diagrams; Load calculation for lighting and power circuits; Sizing of cables and distribution boards; Estimation of materials and labor costs.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

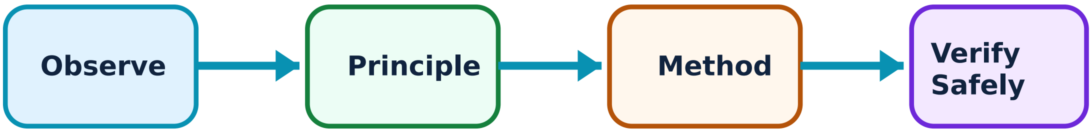
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Estimation and Costing for Domestic Wiring: identify the system, state its principle, follow the correct method and verify the result safely.

1. Layout and Load Calculation−

Estimation starts with a building plan. A wiring layout shows the position of points, switches, and the distribution board. Total load is calculated by summing the wattage of all appliances, which determines the sizing of the main switch and service mains.

Study and application method

Prepare a 'Wiring Layout' for a two-room house conceptually; calculate the 'Total Connected Load' for a given list of appliances.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Prepare a 'Wiring Layout' for a two-room house conceptually; calculate the 'Total Connected Load' for a given list of appliances.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Underestimating the load can cause frequent tripping or equipment damage. Do not design distribution systems without authorised diversity factor and future-expansion analysis.

2. Material and Cost Estimation−

A material list includes cables, conduits, switches, MCBs, and fixing accessories. Costing involves multiplying the quantities by current market rates and adding labor charges (often per point or per meter). Contingency charges are added for unforeseen expenses.

Study and application method

Prepare a 'Bill of Materials' for a simple domestic installation; explain how 'Labor Costs' are estimated for wiring projects.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Prepare a 'Bill of Materials' for a simple domestic installation; explain how 'Labor Costs' are estimated for wiring projects.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Material prices fluctuate. Do not finalize project budgets without authorised verification of the latest market rates and tax implications.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Installation of Distribution and Earthing

Installation of Main Switch and Energy Meter; Distribution Board (DB) arrangement; Sub-circuits: Lighting and Power; Earthing systems: Pipe and Plate earthing; Earth resistance measurement.

Approx. 16 hours | CO3 | Understand · Apply

Learning goals

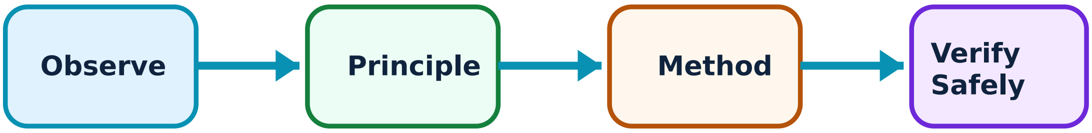
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Installation of Distribution and Earthing: identify the system, state its principle, follow the correct method and verify the result safely.

1. Distribution Board and Sub-circuits−

The Distribution Board (DB) divides the main supply into sub-circuits. Lighting circuits handle low-power loads, while power circuits handle heavy appliances like ACs and geysers. Protective devices like MCBs (Miniature Circuit Breakers) and RCCBs (Residual Current Circuit Breakers) are installed in the DB.

Study and application method

Sketch the internal wiring of a 'Distribution Board' showing MCBs and Neutral link; differentiate between 'Lighting' and 'Power' sub-circuits.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the internal wiring of a 'Distribution Board' showing MCBs and Neutral link; differentiate between 'Lighting' and 'Power' sub-circuits.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Improper sub-circuit division can overload the neutral wire. Do not wire distribution boards without authorised balancing of phases and proper neutral sizing.

2. Earthing and Protection−

Earthing provides a low-resistance path for leakage current to flow to the ground, protecting users from electric shock. Pipe earthing is common for residential buildings. The earth resistance should be as low as possible (typically < 5 ohms for domestic).

Study and application method

Describe the procedure for 'Pipe Earthing' with a conceptual diagram; explain the role of an 'RCCB' in preventing electric shock.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the procedure for 'Pipe Earthing' with a conceptual diagram; explain the role of an 'RCCB' in preventing electric shock.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A high-resistance earth connection is ineffective. Do not commission an installation without authorised measurement and verification of the earth resistance.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Testing, Commissioning and Troubleshooting

Testing of new installations: Insulation resistance, Continuity, Polarity, Earth resistance; Commissioning procedures; Common wiring faults and troubleshooting; Basics of home automation and UPS installation.

Approx. 14 hours | **CO4** | **Understand · Apply**

Learning goals

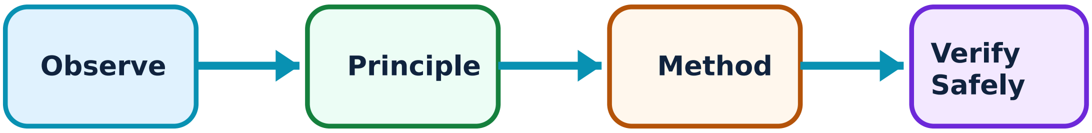
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Testing, Commissioning and Troubleshooting: identify the system, state its principle, follow the correct method and verify the result safely.

1. Installation Testing–

Before energizing, a new installation must be tested. Insulation resistance test (using a Megger) ensures no leakage between conductors. Continuity test verifies the circuit path. Polarity test ensures switches are in the phase (live) wire.

Study and application method

List the four mandatory tests to be performed on a new domestic installation; explain the 'Insulation Resistance Test' conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List the four mandatory tests to be performed on a new domestic installation; explain the 'Insulation Resistance Test' conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Testing must be done on a de-energized system. Do not apply high-voltage test signals (Megger) without authorised disconnection of sensitive electronic equipment.

2. Troubleshooting and Modern Trends−

Common faults include short circuits, open circuits, and earth leaks. Modern residential electrification includes installing UPS/Inverters for backup and home automation systems for smart control of lights and appliances.

Study and application method

Describe how to troubleshoot a 'Short Circuit' in a domestic sub-circuit; explain the benefits of 'Home Automation' conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe how to troubleshoot a 'Short Circuit' in a domestic sub-circuit; explain the benefits of 'Home Automation' conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Improper UPS/Inverter wiring can cause back-feeding into the grid. Do not install backup power systems without authorised change-over switches and proper isolation.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Wiring Systems and Safety Standards

Use the concept flow to revise observation, principle, method and verification.

Module 2: Estimation and Costing for Domestic Wiring

Use the concept flow to revise observation, principle, method and verification.

Module 3: Installation of Distribution and Earthing

Use the concept flow to revise observation, principle, method and verification.

Module 4: Testing, Commissioning and Troubleshooting

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Prepare a wiring layout and single-line diagram for your study room.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify and list the ratings of MCBs and the RCCB in your home distribution board.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Compare PVC and MS conduits in a conceptual table showing three differences.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the total wattage and current for a kitchen with a fridge, microwave, and induction stove.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Observe a professional electrician performing an earth resistance test or wiring a switchboard.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Wiring Systems and Safety Standards

Explain Wiring systems and accessories and state one correct method, application or precaution.−

Electrical wiring involves the systematic installation of cables and accessories. Casing-capping is used for surface wiring, while conduit wiring (surface or concealed) provides better protection. Accessories like modular switches and multi-pin sockets are selected based on current rating and application.

Answer framework: Compare 'Concealed Conduit' and 'Casing-Capping' wiring systems; list five common wiring accessories used in a bedroom.

Important point: Improper cable selection can lead to overheating and fires. Do not install wiring without authorised current-carrying capacity (ampacity) analysis based on the load.

Explain IE Rules and Safety and state one correct method, application or precaution.−

The Indian Electricity (IE) rules mandate safety standards for electrical installations. Key rules include minimum height of switches, distance between light points and fans, and mandatory use of protective devices. The National Electrical Code (NEC) provides comprehensive guidelines for residential safety.

Answer framework: State three 'IE Rules' related to domestic electrical installations; explain the importance of 'Electrical Safety Symbols'.

Important point: Electrical work is hazardous. All domestic installations must follow authorised safety codes and be performed by a certified electrician.

Module 2 — Estimation and Costing for Domestic Wiring

Explain Layout and Load Calculation and state one correct method, application or precaution.−

Estimation starts with a building plan. A wiring layout shows the position of points, switches, and the distribution board. Total load is calculated by summing the wattage of all appliances, which determines the sizing of the main switch and service mains.

Answer framework: Prepare a 'Wiring Layout' for a two-room house conceptually; calculate the 'Total Connected Load' for a given list of appliances.

Important point: Underestimating the load can cause frequent tripping or equipment damage. Do not design distribution systems without authorised diversity factor and future-expansion analysis.

Explain Material and Cost Estimation and state one correct method, application or precaution.−

A material list includes cables, conduits, switches, MCBs, and fixing accessories. Costing involves multiplying the quantities by current market rates and adding labor charges (often per point or per meter). Contingency charges are added for unforeseen expenses.

Answer framework: Prepare a 'Bill of Materials' for a simple domestic installation; explain how 'Labor Costs' are estimated for wiring projects.

Important point: Material prices fluctuate. Do not finalize project budgets without authorised verification of the latest market rates and tax implications.

Module 3 — Installation of Distribution and Earthing

Explain Distribution Board and Sub-circuits and state one correct method, application or precaution.−

The Distribution Board (DB) divides the main supply into sub-circuits. Lighting circuits handle low-power loads, while power circuits handle heavy appliances like ACs and geysers. Protective devices like MCBs (Miniature Circuit Breakers) and RCCBs (Residual Current Circuit Breakers) are installed in the DB.

Answer framework: Sketch the internal wiring of a 'Distribution Board' showing MCBs and Neutral link; differentiate between 'Lighting' and 'Power' sub-circuits.

Important point: Improper sub-circuit division can overload the neutral wire. Do not wire distribution boards without authorised balancing of phases and proper neutral sizing.

Explain Earthing and Protection and state one correct method, application or precaution.−

Earthing provides a low-resistance path for leakage current to flow to the ground, protecting users from electric shock. Pipe earthing is common for residential buildings. The earth resistance should be as low as possible (typically < 5 ohms for domestic).

Answer framework: Describe the procedure for 'Pipe Earthing' with a conceptual diagram; explain the role of an 'RCCB' in preventing electric shock.

Important point: A high-resistance earth connection is ineffective. Do not commission an installation without authorised measurement and verification of the earth resistance.

Module 4 — Testing, Commissioning and Troubleshooting

Explain Installation Testing and state one correct method, application or precaution.−

Before energizing, a new installation must be tested. Insulation resistance test (using a Megger) ensures no leakage between conductors. Continuity test verifies the circuit path. Polarity test ensures switches are in the phase (live) wire.

Answer framework: List the four mandatory tests to be performed on a new domestic installation; explain the 'Insulation Resistance Test' conceptually.

Important point: Testing must be done on a de-energized system. Do not apply high-voltage test signals (Megger) without authorised disconnection of sensitive electronic equipment.

Explain Troubleshooting and Modern Trends and state one correct method, application or precaution.−

Common faults include short circuits, open circuits, and earth leaks. Modern residential electrification includes installing UPS/Inverters for backup and home automation systems for smart control of lights and appliances.

Answer framework: Describe how to troubleshoot a 'Short Circuit' in a domestic sub-circuit; explain the benefits of 'Home Automation' conceptually.

Important point: Improper UPS/Inverter wiring can cause back-feeding into the grid. Do not install backup power systems without authorised change-over switches and proper isolation.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Wiring Systems and Safety Standards.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Estimation and Costing for Domestic Wiring.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Installation of Distribution and Earthing.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Testing, Commissioning and Troubleshooting.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Types of wiring: Cleat, Casing-Capping, Conduit (PVC/MS); Wiring accessories: Switches, sockets, lamp holders, ceiling roses; Indian Electricity (IE) rules for domestic wiring; National Electrical Code (NEC) overview..
2. Develop a complete answer or practical plan for: Preparation of wiring layout plans; Single-line diagrams; Load calculation for lighting and power circuits; Sizing of cables and distribution boards; Estimation of materials and labor costs..
3. Develop a complete answer or practical plan for: Installation of Main Switch and Energy Meter; Distribution Board (DB) arrangement; Sub-circuits: Lighting and Power; Earthing systems: Pipe and Plate earthing; Earth resistance measurement..
4. Develop a complete answer or practical plan for: Testing of new installations: Insulation resistance, Continuity, Polarity, Earth resistance; Commissioning procedures; Common wiring faults and troubleshooting; Basics of home automation and UPS installation..

LAST-MILE REVISION

Quick Revision

Module 1: Wiring Systems and Safety Standards

Types of wiring: Cleat, Casing-Capping, Conduit (PVC/MS); Wiring accessories: Switches, sockets, lamp holders, ceiling roses; Indian Electricity (IE) rules for domestic wiring; National Electrical Code (NEC) overview.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Estimation and Costing for Domestic Wiring

Preparation of wiring layout plans; Single-line diagrams; Load calculation for lighting and power circuits; Sizing of cables and distribution boards; Estimation of materials and labor costs.

- Match each topic to CO2.
- Redraw or rehearse one key method.

- Check one common mistake/precaution.

Module 3: Installation of Distribution and Earthing

Installation of Main Switch and Energy Meter; Distribution Board (DB) arrangement; Sub-circuits: Lighting and Power; Earthing systems: Pipe and Plate earthing; Earth resistance measurement.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Testing, Commissioning and Troubleshooting

Testing of new installations: Insulation resistance, Continuity, Polarity, Earth resistance; Commissioning procedures; Common wiring faults and troubleshooting; Basics of home automation and UPS installation.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Wiring systems and accessories	Electrical wiring involves the systematic installation of cables and accessories.
IE Rules and Safety	The Indian Electricity (IE) rules mandate safety standards for electrical installations.
Layout and Load Calculation	Estimation starts with a building plan.
Material and Cost Estimation	A material list includes cables, conduits, switches, MCBs, and fixing accessories.
Distribution Board and Sub-circuits	The Distribution Board (DB) divides the main supply into sub-circuits.
Earthing and Protection	Earthing provides a low-resistance path for leakage current to flow to the ground, protecting users from electric shock.
Installation Testing	Before energizing, a new installation must be tested.
Troubleshooting and Modern Trends	Common faults include short circuits, open circuits, and earth leaks.

LEARNING RESOURCES

References

Recommended residential electrification references

1. S. L. Uppal and G. C. Garg, Electrical Wiring, Estimating and Costing.
2. J. B. Gupta, A Course in Electrical Installation Estimating and Costing.
3. K. B. Raina and S. K. Bhattacharya, Electrical Design Estimating and Costing.
4. Bureau of Indian Standards, National Electrical Code (NEC).

Approved public residential electrification sources

- https://nptel.ac.in/courses/117106108
- https://onlinecourses.swayam2.ac.in/nou21_ee01/preview

These sources provide the verified study basis while official Course 6034C detail is unavailable; they do not replace official building designs, authorised electrical inspections, certified contractor work or authorised utility plans.